

535/1

PHYSICS

Paper 1

March, 2026

2 Hours 15 Minutes



Pre- registration 2026

S.4 PHYSICS

535/1 Paper 1

INSTRUCTIONS TO CANDIDATE:

- *This paper consists of two sections; A and B. It has five examination items.*
- *Section A has two compulsory items.*
- *Section B has two parts; I and II. Answer one question from part I, part II is compulsory.*
- *Answer **four** items in all.*
- *Any additional item(s) answered will not be scored.*
- *All answers must be written in the booklets provided.*

Item one

The choir of a certain high school went to a studio to record an audio version of their school anthem. While in the studio, students made the following observations:

- The walls of the studio were covered by soft, thick woolen materials.
- The color of their uniforms changed as the different lights flashed in the studio. The uniform which was initially of blue skirts and white shirts with yellow collars, kept changing colors as green and red lights flashed on them.
- There was a section at the studio where two mirrors were inclined at an angle of 60° and when one student went there, she saw multiple images of herself.

They were later hosted at an FM radio station studio to premier their school anthem, but in that area, there are two FM radio stations known as Tower FM broadcasting at ***100.2 MHz*** and KT FM broadcasting at ***90.9 MHz***

All these experiences when narrated to the friends in the lower classes back at school looked mysterious.

Hint:

- ***The speed of electromagnetic waves in air = $3 \times 10^8 \text{ ms}^{-1}$***
- ***The wavelength of the waves from the station they visited = 3.3 m***

Task:

You were one of the choir team who went to the studio, with your knowledge of physics:

- a) Explain to your friends in the lower classes why the walls of the studios are made the way you observed and what would go wrong if they are left bare and hard.
- b) Explain how and why the colors of the students' uniforms kept on changing as the flash lights changed color.
- c) Determine how many images your friend saw while in the studio.
- d) Determine which FM radio station you went to.

Item two

In 2014, Brazil hosted FIFA world cup where different countries participated in the tournament. Some countries such as Uganda did not qualify for the tournament and hence had to watch the matches live on their TV screens. In one of the games, it was evident that it was day time in Brazil yet in Uganda it was night time which puzzled John as he was watching the game live on TV, during the same period of world cup tournament (12th June to 13th July of 2014) it was winter in Brazil which affected some participant nations in the tournament.

Hint: At John's home there was a satellite DSTV dish.

Task:

Using physics knowledge help John to understand;

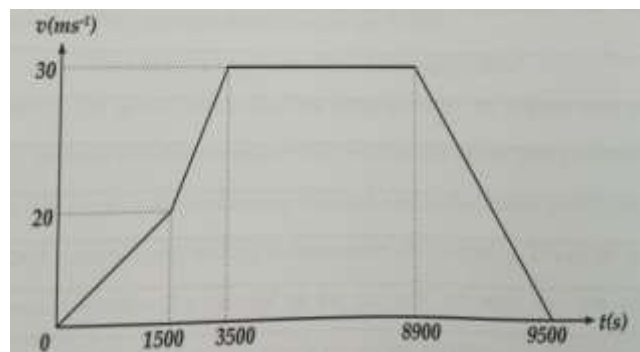
- (a) The cause of night in Uganda yet it was daytime in Brazil
- (b) The cause of variation in seasons
- (c) How he was able to watch the matches in Brazil while in Uganda.

SECTION B

PART I

Item three

The vehicle that was hired to transport students for a study tour was equipped with a computer software that draws the motion of the vehicle from which analysis is made so that refueling is done. Refueling is done based on the distance the vehicle has moved and the driver should refuel every after 210 km using reserve fuel carried in well-sealed jerrycans. The newly employed supervisor of the vehicle is not yet conversant with the interpretation and analysis of this graph to determine whether the vehicle was indeed refueled. The figure below shows the motion of the vehicle.



During the journey, some students were inconvenienced by the forward and back ward jerking whenever the vehicle reduced and increased speed suddenly. The students also wondered why despite the vehicle being closed all the time, the interior remained cold.

Task: Help the students;

- a) Understand why they experienced that kind of movement.
- b) How the car interior was kept cold.
- c) Advise the supervisor whether the vehicle was refueled or not since the fuel was used.
- d) Describe a real life application of each of the concepts identified in the scenario

Item four

In a certain area, a farmer's goat fell in an underground water tank pit that was left uncovered. A worker in the farm has been tasked to remove the goat from the pit but the goat is too heavy to be removed by carrying it on the ladder. The worker has been provided with two pulleys and a long strong string among other materials to help in lifting the goat from the pit. The string will burn and break if the temperature at the contact points exceeds 59°C .

Hint: All the work done in lifting the goat from the pit transforms to heat energy at the contact points.

Acceleration due to gravity = 10 ms^{-2}

Heat capacity of the material of the string = 102 JK^{-1}

The pit is 5.5 m deep.

Mass of the goat = 65 kg and the initial temperature at the contact points is expected to 22°C

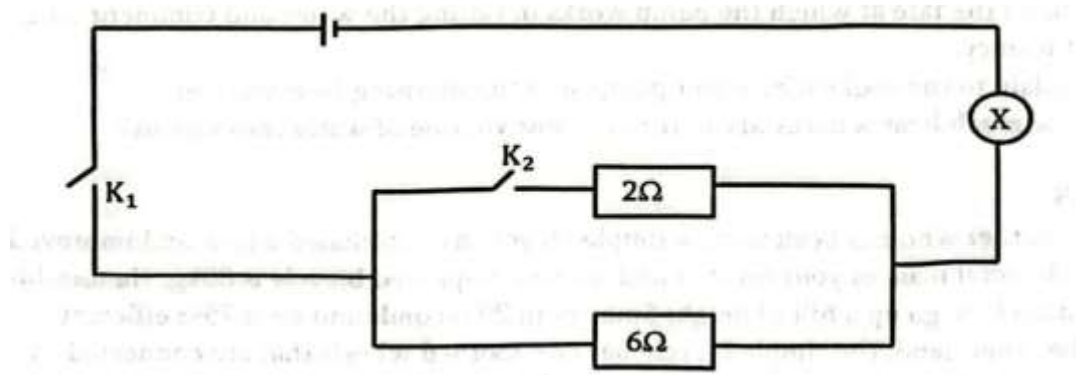
Task: Help the worker;

- a) With an appropriate design of a system that will help him to remove the goat from the pit.
- b) Why some heat will be generated and how it can be minimized.
- c) To determine whether the material of the strings is appropriate for use.

PART II

Item five

A group of learners have been provided with 2 switches, 1 dry cell of emf 10 V and negligible internal resistance, two resistors of resistance $6\ \Omega$ and $2\ \Omega$, a steel bar and a bar magnet . They connected a circuit as shown in the figure below;



However, they are displeased with having two switches in the circuit. The learners are wondering if they could have only one switch which when closed can produce a large current that flows through the Ammeter connected at position X. The learners are also disturbed on which switch should be closed and which one should be opened for maximum current to flow through the ammeter at position X.

Task:

Using your knowledge of physics;

- (a) Help these learners to clear their wonders.
- (b) explain how the Ammeter reading would change if;
 - (i) The temperature of the connecting wires is increased.
 - (ii) Similar connecting wires of large cross sectional area were used.
- (c) Help this group to briefly describe how the steel bar can be magnetized using the given bar magnet clearly explaining the polarity of the made magnet.

END

“If you can dream it, you can do it”